TEXORA · LECTURE SHEET

Kinetic Theory of Gases

How pressure, temperature and heat emerge from molecular motion: the kinetic model and its assumptions, the Maxwell speed distribution, degrees of freedom and equipartition, specific heats, mean free path and transport, and the van der Waals correction for real gases.

PHY-1203 · PHYSICS-II · TOPIC 3 · LEVEL 1 TERM 2

1. The kinetic model — assumptions

Kinetic theory explains the gas laws by picturing a gas as a swarm of molecules in ceaseless random motion. The **ideal-gas model** assumes:

- Molecules are point masses of negligible volume compared with the container.
- No forces act between molecules except during collisions; between collisions they move in straight lines at constant speed.
- All collisions (with walls and each other) are perfectly elastic.
- The container holds so many molecules that averages are meaningful and the distribution of velocities is isotropic (no preferred direction).

Counting the momentum delivered to a wall by this swarm (each molecule of mass m with x-velocity u delivers 2mu per hit, and returns after time 2l/u for a box of side l) gives the central result:

$$p = (1/3)\,\rho\,\bar{c}^2 = (1/3)\,n\,m\,\langle c^2\rangle$$

where $\bar{c}^2$ is the **mean-square speed**. Comparing with pV = nRT identifies temperature with molecular kinetic energy:

$$(1/2)\,m\,\langle c^2\rangle = (3/2)\,kT, \quad \text{so} \quad (3/2)\,RT \text{ per mole}$$

Temperature *is* average translational kinetic energy, in disguise — at 300 K each molecule carries $\tfrac{1}{2}m\langle c^2\rangle = 1.5 \times 1.381 \times 10^{-23} \times 300 \approx \mathbf{6.2 \times 10^{-21}\ J}$, and each mole (3/2)RT ≈ 3.74 kJ.

2. The three characteristic speeds

Molecules do not all move at one speed: Maxwell’s distribution spreads them in a skewed bell whose peak shifts right and flattens as T rises. Three speeds summarise it (M = molar mass):

SPEED	FORMULA	FOR O_2 AT 300 K (M = 0.032 KG MOL^{-1})
Most probable c_p	√(2RT/M)	≈ 395 m s^{-1}
Mean c̄	√(8RT/πM)	≈ 446 m s^{-1}
Root-mean-square c_rms	√(3RT/M)	≈ 484 m s^{-1}

The fixed ratio c_p : c̄ : c_rms = 1 : 1.128 : 1.225 holds for every ideal gas. Because c_rms ∝ √(T/M), light gases move fast (H_2 at 300 K: ≈ 1,930 m s^{-1} — comparable to Earth's escape processes, which is why the atmosphere keeps almost no hydrogen) and heating a gas from 300 K to 1,200 K only doubles molecular speeds.

3. Degrees of freedom and equipartition

A **degree of freedom** is an independent quadratic term in a molecule's energy. The **equipartition theorem** assigns (1/2)kT of average energy to each. Counting them predicts the molar heat capacities exactly:

GAS	F	C_V = (F/2)R	C_P = C_V + R	Γ = C_P/C_V = 1 + 2/F
Monatomic (He, Ar)	3 (translation)	12.5 J mol^{-1} K^{-1}	20.8	1.67
Diatomic (N_2, O_2)	5 (+2 rotation)	20.8	29.1	1.40
Non-linear polyatomic (H_2O)	6 (+3 rotation)	24.9	33.2	1.33

Measured values match at room temperature. The model's famous failure — vibration contributes nothing at ordinary temperatures, though classical physics says it should — was an early signpost to quantum theory, where vibrational levels are spaced wider than kT and stay "frozen".

4. Mean free path and transport phenomena

A molecule travels a jagged path, colliding billions of times per second. The average step between collisions is the **mean free path**

$$\lambda = 1/(\sqrt{2}\,\pi\, d^2\, n)$$

(d = molecular diameter, n = number density; √2 corrects for the targets also moving). For air at STP (d ≈ 3.7 × 10^{-10} m, n = 2.69 × 10^{25} m^{-3} — Loschmidt's number), λ ≈ **6 × 10^{-8} m**, about 160 molecular diameters, and the collision frequency c̄/λ ≈ 446/(6 × 10^{-8}) ≈ 7 × 10^{9} s^{-1}. Transport coefficients follow

from molecules carrying properties across one free path: viscosity $\eta = (1/3)\rho\bar{c}\lambda$, thermal conductivity of the same form times the specific heat, diffusion $D = (1/3)\bar{c}\lambda$. A classic triumph: η from the formula is **independent of pressure** (halving density doubles λ, cancelling out) — a strange prediction Maxwell verified experimentally before believing it himself. It fails only at very low pressure, where λ reaches the vessel size.

5. Real gases — the van der Waals equation

Real gases deviate because molecules *do* attract each other slightly and *do* occupy volume. Van der Waals patched both:

$$(p + a/V^2)(V - b) = RT \quad \text{(per mole)}$$

b subtracts the molecules' own volume ($\approx 4 \times$ molecular volume per mole); **a/V^2** adds back the pressure lost to inward attraction at the walls. The equation predicts a **critical point** above which no pressure liquefies the gas, with constants in terms of a and b:

$$V_c = 3b, \quad p_c = a/(27b^2), \quad T_c = 8a/(27Rb)$$

WORKED EXAMPLE — CO_2

$a = 0.364\ \text{Pa·m}^6\ \text{mol}^{-2}$, $b = 4.27 \times 10^{-5}\ \text{m}^3\ \text{mol}^{-1}$.
$T_c = 8 \times 0.364/(27 \times 8.314 \times 4.27 \times 10^{-5}) = 2.912/(9.59 \times 10^{-3}) \approx$ **304 K** — measured: 304.1 K.
$p_c = 0.364/(27 \times (4.27 \times 10^{-5})^2) \approx 7.4 \times 10^6$ Pa ≈ **73 atm** — measured: 73.8 atm. A two-parameter model from 1873 landing within a percent: that is why it is still taught.

6. Key points

- $p = (1/3)\rho\langle c^2\rangle$ and $(1/2)m\langle c^2\rangle = (3/2)kT$: pressure and temperature are molecular bookkeeping.
- Speeds: $c_p < \bar{c} < c_{rms}$ in the ratio 1 : 1.128 : 1.225; all $\propto \sqrt{(T/M)}$.
- Equipartition: $(1/2)kT$ per quadratic degree of freedom; $C_V = (f/2)R$; $\gamma = 1 + 2/f$ (1.67 / 1.40 / 1.33 for f = 3 / 5 / 6).
- $\lambda = 1/(\sqrt{2}\pi d^2 n) \approx 6 \times 10^{-8}$ m for air at STP; viscosity $\eta = (1/3)\rho\bar{c}\lambda$, independent of pressure.
- Van der Waals: a corrects attraction, b corrects volume; critical constants $V_c = 3b$, $p_c = a/27b^2$, $T_c = 8a/27Rb$.

7. Practice

Answer in your own words; brief model answers follow.

1. Calculate the rms speed of nitrogen molecules ($M = 0.028$ kg mol^{-1}) at 300 K and at 600 K.
2. A gas's temperature is raised at constant volume until its pressure triples. By what factor do (a) the mean-square speed, (b) the rms speed, change?
3. Show that γ for a gas with $f = 7$ degrees of freedom is 1.29, and state C_V.
4. Estimate the mean free path of air at 100 Pa (other data as in this sheet, STP $n = 2.69 \times 10^{25}$ m^{-3} at 10^5 Pa). What happens to gas viscosity as pressure falls toward this value?
5. For oxygen ($a = 0.138$ Pa·m^6 mol^{-2}, $b = 3.18 \times 10^{-5}$ m^3 mol^{-1}), compute T_c and compare with the measured 154.6 K.
6. Why does the ideal-gas model fail at high pressure even if the temperature is high? Answer through the two van der Waals corrections.

Answer key

1. $c_{rms} = \sqrt{(3RT/M)} = \sqrt{(3 \times 8.314 \times 300/0.028)} = \sqrt{267{,}236} \approx 517$ m s^{-1}. At 600 K speed scales by $\sqrt{2}$: ≈ 731 m s^{-1}.
2. Pressure $\propto T$ at fixed V, so T triples. (a) $\langle c^2 \rangle \propto T \rightarrow \times 3$. (b) $c_{rms} \propto \sqrt{T} \rightarrow \times\sqrt{3} \approx 1.73$.
3. $\gamma = 1 + 2/f = 1 + 2/7 = 1.286 \approx 1.29$. $C_V = (f/2)R = 3.5 \times 8.314 \approx 29.1$ J mol^{-1} K^{-1}.
4. n scales with pressure: $n = 2.69 \times 10^{25} \times (100/10^5) = 2.69 \times 10^{22}$ m^{-3}. $\lambda = 6 \times 10^{-8} \times (10^5/100) = 6 \times 10^{-5}$ m = 0.06 mm. Viscosity stays constant while $\lambda \ll$ vessel size ($\rho\lambda$ constant), but once λ approaches the container dimensions the formula breaks and apparent viscosity falls — the regime of vacuum technology and Pirani gauges.
5. $T_c = 8a/(27Rb) = 8 \times 0.138/(27 \times 8.314 \times 3.18 \times 10^{-5}) = 1.104/(7.14 \times 10^{-3}) \approx 154.7$ K — essentially exact against 154.6 K measured.
6. At high pressure the free volume shrinks toward the molecules' own volume, so treating them as points (V instead of $V - b$) badly overstates the space available; simultaneously the density is high enough that intermolecular attraction (the a/V^2 term) measurably softens wall impacts. High temperature fixes only the second problem partially — speeds rise but the volume error remains, so the b correction matters at any temperature once the gas is dense.

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